

Higher Structures in Rome
17-18 Sept. 2026

GOOD COMPLETION for
KOSZUL OPERADS

Francesca Pratioli
Utrecht University

joint w/ Maite Carli &
Gijs Heuts

Bor-Cobor duality

$$\text{Alg}(\mathcal{C})^{\text{aug}} \begin{array}{c} \xrightarrow{\text{Bor}} \\ \xleftarrow{\text{Cobor}} \end{array} \text{coAlg}(\mathcal{C})^{\text{aug}}$$

Bor-Coban duality $\text{Alg}(\mathcal{C})^{\text{aug}} \xrightleftharpoons[\text{Coban}]{\text{Bor}} \text{coAlg}(\mathcal{C})^{\text{coaug}}$

Rational homotopy theory

Quillen:

$\text{Lie}(D(\mathbb{Q})) \xrightleftharpoons[\text{prim}]{\text{CE}} \text{coCAlg}(D(\mathbb{Q}))$

Bor-Cobon duality $\text{Alg}(C)^{\text{aug}} \xrightleftharpoons[\text{Cobon}]{\text{Bor}} \text{coAlg}(C)^{\text{aug}}$

Rational homotopy theory

Quillen:

$\text{Lie}(D(Q))^{\geq 1} \xrightleftharpoons[\text{prim}]{\text{CE}} \text{coCAlg}(D(Q))^{\geq 2}$

instances of
OPERADIC KOSZUL DUALITY

category of derived k -modules.

Object in $D(k) \rightsquigarrow B\theta$ cooperad in $D(k)$
related by an adjunction

$$\text{Alg}_{\theta}(D(k))^{\text{aug}} \begin{array}{c} \xrightarrow{\text{indec}_{\theta}} \\ \perp \\ \xleftarrow{\text{prim}_{B\theta}} \end{array} \text{coAlg}_{B\theta}(D(k))^{\text{aug}}$$

category of derived k -modules.

$\mathcal{O}$ operad in $D(k) \rightsquigarrow B\mathcal{O}$ cooperad in $D(k)$
related by an adjunction

$$\boxed{\text{Alg}_{\mathcal{O}}(D(k))^{\text{aug}} \begin{array}{c} \xrightarrow{\text{indec}_{\mathcal{O}}} \\ \perp \\ \xleftarrow{\text{prim}_{B\mathcal{O}}} \end{array} \text{coAlg}_{B\mathcal{O}}(D(k))^{\text{aug}}}$$

, Bor-Cobar duality: $\mathcal{O} = \text{Ass}$, $B\text{Ass} = s\text{Ass}^{\vee}$

, Notionel Htheory: $\mathcal{O} = \text{Lie}$, $B\text{Lie} = s\text{Com}^{\vee}$

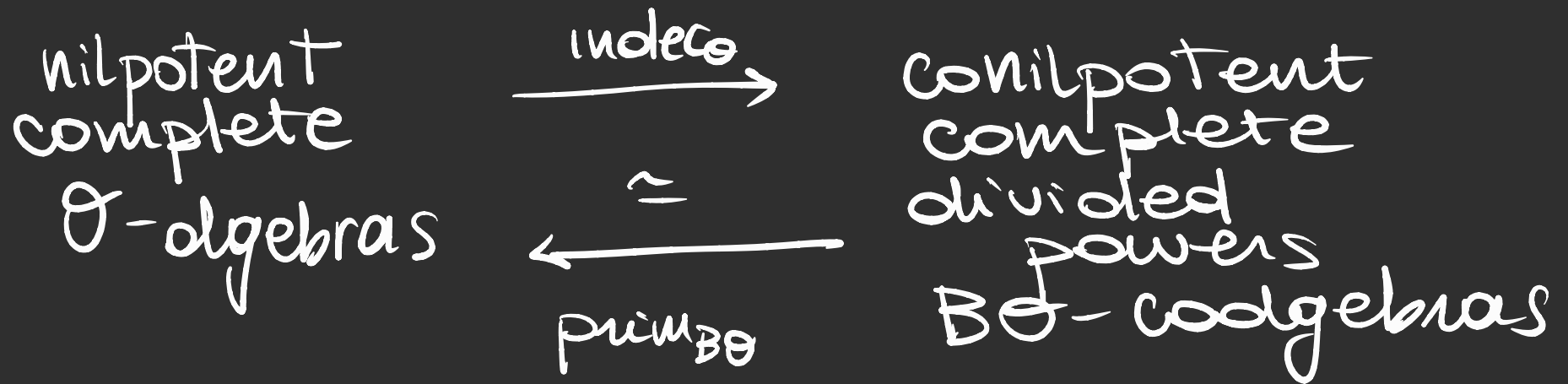
We will be interested in

$\mathcal{O} \simeq \mathbb{H}_n$ k -LINEAR LITTLE n -DISKS
OPERAD

$$\mathbb{H}_n(m) = C_* \left(\mathbb{H}_n^{\text{top}}(m) ; k \right) \in D(k)$$

[Ching - Salvatore] $\text{indec}_{\mathbb{H}_n} \simeq \text{Bar}^{(n)}$

Thm [Heuts] The Koszul adjunction
restricts to an equivalence



Moreover, these subcategories are maximal.

for $X \in \mathbf{Algo}$
NILCOMPLETION

$$X \xrightarrow{\sim} \varprojlim_n t_n X$$

for $X \in \mathcal{A}lgo$

NILCOMPLETION

$$X \xrightarrow{\sim} \varprojlim_n t_n X$$

HOMOLOGICAL
COMPLETION

[Bousfield-Kan]

[Francis-Gaitsgory]

for $X \in \mathcal{A}lge$

NILCOMPLETION

$$X \xrightarrow{\sim} \varprojlim_n t_n X$$

HOMOLOGICAL
COMPLETION
[Bousfield-Kan]
[Francis-Gaitsgory]

Def [Heuts] $\mathcal{O}$ has **GOOD COMPLETION** if
every trivial $\mathcal{O}$ -algebra is nilcomplete.

Prop. If Θ has GOOD completion

for $X \in \text{Algo}$

NILCOMPLETION



HOMOLOGICAL
COMPLETION

Prop. If Θ has GOOD COMPLETION

for $X \in \text{Algo}$

NILCOMPLETION $\longleftrightarrow$ HOMOLOGICAL COMPLETION

Thm [Carli-Hents-P.] $\text{chark} = 0$.

Let Θ be a BINARY QUADRATIC KOSZUL operad. Then Θ has good completion.

Prop. If Θ has GOOD COMPLETION

for $X \in \text{Algo}$

NILCOMPLETION $\longleftrightarrow$ HOMOLOGICAL COMPLETION

Thm [Carli-Hents-P.] $\text{chark} = 0$.

Let Θ be a BINARY QUADRATIC KOSZUL operad. Then Θ has good completion.

Corollary The k -linear $\mathbb{H}_n$ operad has good completion for $1 \leq n \leq \infty$

2.1. Setting

$(\mathcal{C}, \otimes)$ stable s.m. ∞ categ., $-\otimes-$ colim-pres.

e.g. • $(D(K), \otimes)$, $H_0(D(K)) = \text{Ch}_K[\mathbf{q}|\mathbf{s}]$

• $(Sp, \wedge)$ $H_0(Sp)$ the stable
homotopy cat.

> NON-UNITAL and REDUCED sym. seqs. in ℓ
 $H(0)=0$ $H(1)=1$

$$S\text{Seq}(\ell) := \prod_{n \geq 2} \ell^{B\mathbb{Z}_n}$$

Monoidal category with $\text{---} \circ \text{---}$, unit 11
 $11(n) = \begin{cases} 1 & n=1 \\ 0 & n \neq 1 \end{cases}$

> NON-UNITAL and REDUCED sym. seqs. in $\mathcal{C}$
 $H(0)=0$ $H(1)=1$

$$S\text{Seq}(\mathcal{C}) := \prod_{n \geq 2} \mathcal{C}^{\mathbb{B}\Sigma_n}$$

Monoidal category w/ht $\text{---} \circ \text{---}$, unit 1
 $1(n) = \begin{cases} 1 & n=1 \\ 0 & n \neq 1 \end{cases}$

$$Op(\mathcal{C}) := \text{Alg}^{(nu, red)}(S\text{Seq}(\mathcal{C}), \circ)$$

$$coOp(\mathcal{C}) := co\text{Alg}^{(nu, red)}(S\text{Seq}(\mathcal{C}), \circ)$$

> $\mathcal{O}$ -operad. An $\mathcal{O}$ -algebra is $X \in \mathcal{L}$ w/ action maps

$$(\mathcal{O}(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+} \longrightarrow X \quad \forall n \geq 1$$

> $\text{Alg}_0(\mathcal{C}) := \text{Alg}_{\text{sym}_{\mathcal{O}}}(\mathcal{C})$

monad

> $\mathcal{O}$ operad. An $\mathcal{O}$ -algebra is $X \in \mathcal{L}$ w/ action maps

$$(\mathcal{O}(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+} \longrightarrow X \quad \forall n \geq 1$$

> $\text{Alg}_{\mathcal{O}}(C) := \text{Alg}_{\text{sym}_{\mathcal{O}}}(C)$

monad

Rmk $\mathcal{O}$ reduced w/ units, $\mathcal{O}^{un} \subseteq \mathcal{O}$ max. non-un. subopd

$$\text{Alg}_{\mathcal{O}^{un}}(C) \simeq \text{Alg}_{\mathcal{O}}(C)^{\text{aug}/\mathcal{O}(0)}$$

> $\mathbb{Q}$ coperiod. A $\mathbb{Q}$ -coalgebra is $X \in \mathcal{C}$

$$\sim / \quad X \longrightarrow (Q(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+} \quad \forall n$$

> A divided power $\mathbb{Q}$ -coalgebra is

$$X \longrightarrow (Q(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+} \xrightarrow{\text{Nm}} (Q(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+}^{\wedge} \quad \forall n$$

> A ind-comilpotent dp $\mathbb{Q}$ -coalgebra is

$$X \longrightarrow \bigoplus_{n \geq 1} (Q(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+} \longrightarrow \prod_{n \geq 1} (Q(n) \otimes X^{\otimes n})_{n \in \mathbb{Z}_+}$$

Periodic Koszul Duality

$$11 \xrightarrow{i} \theta \xrightarrow{t} 11$$

$$\mathcal{C} \begin{array}{c} \xrightarrow{\text{free}_\theta} \\ \xleftarrow{\text{fg}_\theta} \end{array} \text{Alg}_\theta(\mathcal{C}) \begin{array}{c} \xrightarrow{\text{coto}_\theta} \\ \xleftarrow{\text{triv}_\theta} \end{array} \mathcal{C}$$

3 Periodic Koszul Duality

$$11 \xrightarrow{i} \theta \xrightarrow{t} 11$$

$$\mathcal{C} \begin{array}{c} \xrightarrow{\text{free}_\theta} \\ \xleftarrow{\text{fg}_\theta} \end{array}$$

$$\text{Alg}_\theta(\mathcal{C}) \begin{array}{c} \xrightarrow{\text{coto}} \\ \xleftarrow{\text{triv}_\theta} \end{array} \mathcal{C}$$

coto : COTANGENT FIBER

triv_θ : TRIVIAL ALGEBRA FUNCTOR

$$\mathrm{Alg}_{\mathcal{O}}(C) \begin{array}{c} \xrightarrow{\mathrm{coto}} \\ \xleftarrow{\mathrm{triv}_{\mathcal{O}}} \end{array} C$$

Example:

$$\mathcal{O} = \mathbb{H}_{\infty}^{\mathrm{nu}}, \text{ under } \mathrm{CAlg}^{\mathrm{nu}}(C) \simeq \mathrm{CAlg}(C)^{\mathrm{aug}}$$

$$\bullet \mathrm{Triv}_{\mathcal{O}}(M) \simeq M[\varepsilon]/(\varepsilon^2) \quad \text{square-zero extension}$$

$$\bullet \mathrm{coto}(A \rightarrow K) \simeq \underbrace{\coprod_{A/K}}_{\text{topological cotangent complex}} \bigotimes_A K \quad \text{TAQ}$$

core factors as

$$\text{Alg}_0(\ell) \xrightarrow{\text{indec}_0} \text{coAlg}_{\text{BO}}^{\text{dp}}(\ell) \xrightarrow{\text{fgt}_0} \ell$$

$\rightsquigarrow$

$$\begin{array}{ccc} \text{Alg}_0(\ell) & \xrightarrow{\text{indec}_0} & \text{coAlg}_{\text{BO}}^{\text{dp}}(\ell) \\ & \xleftarrow[\text{prim}_{\text{BO}}]{+} & \end{array}$$

Nilpotent complete algebras

θ opd, $\tau_n \theta$ n -truncation

$$\tau_n \theta(k) = \begin{cases} \theta(k) & k \leq n \\ 0 & \text{else} \end{cases}$$

Tower $\theta \rightarrow \dots \rightarrow \tau_n \theta \rightarrow \tau_{n-1} \theta \rightarrow \dots \rightarrow \tau_1 \theta$

$$\rightsquigarrow \text{Alg}_\theta(C) \begin{matrix} \xrightarrow{\tau_n} \\ \xleftarrow{\quad} \end{matrix} \text{Alg}_{\tau_n \theta}(C)$$

for $X \in \text{Alg}_\theta$, $X \longrightarrow \tau_n X$ unit

Tower $X \rightarrow \dots \rightarrow t_n X \rightarrow t_{n-1} X \rightarrow \dots \rightarrow t_1 X$

Def X is NILCOMPLETE if the natural map
 $X \longrightarrow \varprojlim_n t_n X$ is an equivalence

Tower $X \rightarrow \dots \rightarrow t_n X \rightarrow t_{n-1} X \rightarrow \dots \rightarrow t_1 X$

Def X is NILCOMPLETE if the natural map
 $X \longrightarrow \varprojlim_n t_n X$ is an equivalence

E.g $\mathcal{O} = \mathbb{H}_\infty^{nu}$, $C = D(K)$

$$X = K[x_1, \dots, x_n, \dots] \longrightarrow \varprojlim_n t_n X = K[[x_1, \dots, x_n, \dots]]$$

Thm [Heuts] Koszul duality $(\text{indeco}, \text{prim}_{\text{BO}})$
 restricts to an equivalence

$$\text{Alg}_{\theta}(C)^{(\text{cpl})} \begin{array}{c} \xrightarrow{\text{indeco}} \\ \sim \\ \xleftarrow{\text{prim}_{\text{BO}}} \end{array} \text{coAlg}^{\text{dp}}(C)^{(\text{cpl})}.$$

These are the maximal possible subcats:

$$\forall X \in \text{Alg}_{\theta}(C), \text{ there is an eq. } \begin{array}{c} X \xrightarrow{\quad} \text{prim}_{\text{BO}} \text{indeco } X \\ \searrow \\ \text{here } \begin{array}{c} \text{!} \\ \text{!} \\ \text{!} \end{array} \text{tn } X \\ \xleftarrow{n} \end{array}$$

and dually for $\forall$ BO-coalg.

Homological completion:

[Francis-Gaitsgory] coto factors as

$$\mathrm{Alg}_0(C) \begin{array}{c} \xrightarrow{\mathrm{ind}_{\mathcal{C}_0}^{\mathrm{nil}}} \\ \perp \\ \xleftarrow{\mathrm{prim}_{\mathcal{B}_0}^{\mathrm{nil}}} \end{array} \mathrm{coAlg}_{\mathcal{B}_0}^{\mathrm{dp, nil}}(C) \xrightarrow{\mathrm{fgTo}} \mathcal{C}$$

ind-conilpotent

Homological completion:

[Francis-Gaitsgory] coto factors as

$$\mathrm{Alg}(C) \begin{array}{c} \xrightarrow{\mathrm{indec}_0^{\mathrm{nil}}} \\ \perp \\ \xleftarrow{\mathrm{prim}_{B_0}^{\mathrm{nil}}} \end{array} \mathrm{coAlg}_{B_0}^{\mathrm{dp, nil}}(C) \xrightarrow{\mathrm{fgTo}} \mathcal{C}$$

ind-conilpotent

Def X is homologically complete if

unit $X \rightarrow \mathrm{prim}_{B_0}^{\mathrm{nil}} \mathrm{indec}_0^{\mathrm{nil}} X$ is an equiv.

Good completion

Def An operad $\mathcal{O}$ has GOOD COMPLETION if every trivial $\mathcal{O}$ -algebra is nilcomplete

$$\text{i.e. } \text{Triv}_{\mathcal{O}} \simeq \varprojlim_n \text{Triv}_{\mathcal{O}}$$

$$\text{Rank } \text{triv}_\theta X \simeq \underbrace{B(\tau_n \theta, \theta, 1)}_{=: K_n}(X)$$

$$\rightsquigarrow \text{triv}_\theta X \longrightarrow \varprojlim_n \bigoplus_{d \geq 1} (K_n(d) \otimes X^{\otimes d})_{n \in \mathbb{Z}}$$

$$\varprojlim_n K_n(d) \simeq 0 \text{ if } d \neq 1, \text{ BUT}$$

$\varprojlim_n$ and $\bigoplus_{d \geq 1}$ do NOT commute in general !!

Prop [Herts] $\mathcal{O}$ has good completion

$\Rightarrow$ nilcompletion $\simeq$ homological
completion

Prop [Herts] $\mathcal{O}$ has good completion

$\Rightarrow$ nilcompletion $\simeq$ homological completion

Pf. > Prove it for $t_n X \xrightarrow{\sim} \text{Tot } t_n (t_{n+1} \text{cota})^{\bullet+1} X$

> Use commutative diagram

$$\begin{array}{ccc}
 X & \longrightarrow & \varprojlim_n t_n X \\
 \downarrow & & \downarrow \\
 \text{Tot} (t_{n+1} \text{cota})^{\bullet+1} X & \longrightarrow & \varprojlim_n \text{Tot} (t_n (t_{n+1} \text{cota})^{\bullet+1} X)
 \end{array}$$

$\square$

Operads w/ good completion

- Truncated operads

- $C = D(k)$ $\text{char } k = 0$, $\Theta = \text{Lie}^{nu}$, $\#_1^{nu}$, $\#_\infty^{nu}$

more gen Θ s.t. $H_*(O(d); k)$ in $\text{deg } 0$
 $H_*(BO(d); k)$ in $\text{deg } d-1$

Operads w/ good completion

- Truncated operads
- $C = D(k)$ $\text{char } k = 0$, $\Theta = \text{Lie}^{n_2}$, $\#_1^{n_2}$, $\#_\infty^{n_2}$

more gen Θ s.t. $H_*(O(d); k)$ in $\text{deg } 0$
 $H_*(BO(d); k)$ in $\text{deg } d-1$

NON-EXAMPLE: [Petersen-Roca; Kuo-Yalin]

$\Theta = \text{Free}(s^{-1} \{ k\mu_n \}_{n \geq 1})$ does
NOT have good completion!

Thm [Carli-Hents-P.]

$C = D(K)$, $\text{char } K = 0$.

If $\mathcal{O}$ is a BINARY QUADRATIC operad
and is KOSZUL

then $\mathcal{O}$ has good completion.

Thm [Carli-Hents-P.]

$C = D(K)$, $\text{char } K = 0$.

Yf $\mathcal{O}$ is a BINARY QUADRATIC operad

and is KOSZUL $\mathcal{O}^i \xrightarrow{\sim} B\mathcal{O}$

then $\mathcal{O}$ has good completion.

$$\mathcal{O} \simeq \text{Free}(E(2)) / (R)$$

R of weight 2

Corollary The k -linear $\mathbb{H}_n^{nu}$ -operad in $D(K)$
has good completion

Proof By Tamarkin ($n=2$) & Fresse-Willwacher,
($n \geq 3$),
 $\mathbb{H}_n$ is formal over $\mathbb{Q}$, and

$$\mathbb{H}_n \simeq H_*(\mathbb{H}_n; K) \simeq \text{Pois}_n$$



Proof sketch [Theorem]

- Recall $K_n = B(\tau_n \theta, \theta, 11)$ sym. seq
- We prove the STRONGER CONDITION that
 $(*) \left(K_{n+d}(d) \xrightarrow{g_n} K_n(d) \right)^\vee$ is zero for $d \neq 1$

$$\boxed{1} \quad L_1 \xrightarrow{f_1} L_2 \xrightarrow{f_2} \dots \rightarrow L_n \rightarrow \dots \quad \varprojlim_n L_n = 11$$

tower of right P-modules

$f_i = 0$ outside $d=1$ $\iff$ SS collapses at t_2 -page

For us, $L_n = K_n^\vee$, $P = \mathbb{B}O^\vee$.

[2] E_1 -page is $E_1^{\bullet, n} = H_\bullet(BP(n) \circ P, d_{int}; K)$

[3] θ bin quack $Kos_2 \Rightarrow P$ as well, and

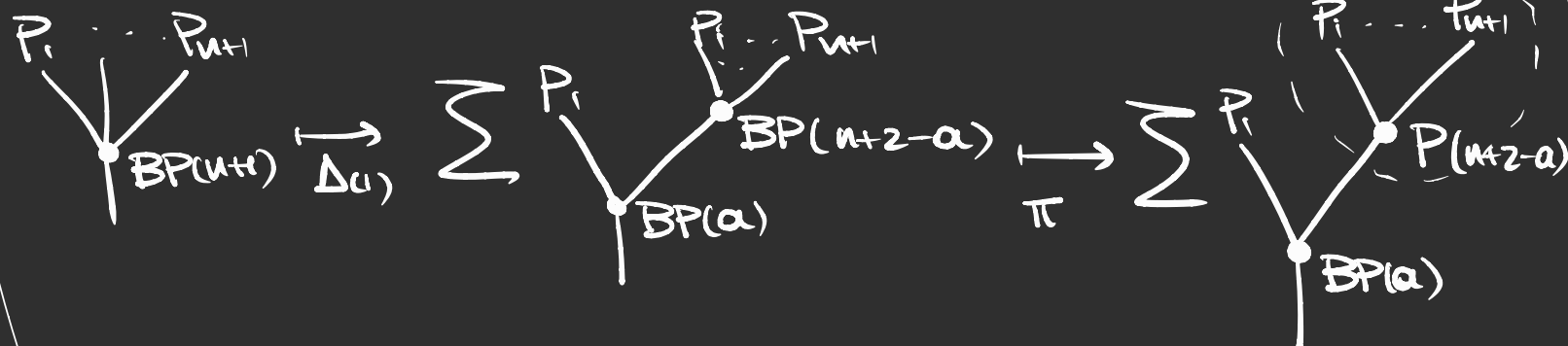
$$d_1^{*, n+1} : H_\bullet(BP(n+1) \circ P) \longrightarrow H_\bullet(BP(n) \circ P)$$

$$d_1^{*, n+1} = H_\bullet(\overline{d\pi})$$

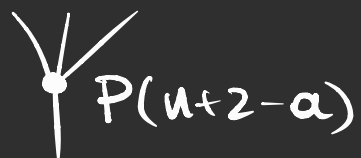
universal
twisting differential
 $\pi: BP \longrightarrow P$

REASON why:

$$d_{\pi} : BP(n+1) \circ P \longrightarrow \bigoplus_{a \leq n} BP(a) \circ P$$



P binary and $a \neq n$

$\Rightarrow$  is a boundary
for $d_{BP \circ P}$ internal

4 P quasiholmic Koszul

$$\triangleleft \Rightarrow P^i \xrightarrow{\sim} BP \quad \text{quasiso}$$

Koszul
dual
cooperad

$$\triangleleft \Rightarrow (P^i \circ P, d_K) \xrightarrow{\sim} (BP \circ P, d_{int} + \overline{d}_\pi) \quad \text{quasiso}$$

for particular

$$\begin{aligned} E_2^{*,*} &= H_*(H_*(BP \circ P, d_{int}), H_*(\overline{d}_\pi)) \\ &\simeq H_{*+*}(BP \circ P, d_{int} + \overline{d}_\pi) \oplus \bigoplus_{Thm} 0 \end{aligned}$$



> What about STABLE $\mathbb{H}_n^{nu}$ period?

Thanks for the
attention!